

MRFD Standard Operational Guideline

SOG #: 1270.030

Topic: Response Time Data Thresholds

Scope: This guideline applies to employees of the Missoula Rural Fire District engaged in response time data acquisition, analysis, and reporting

Purpose: The purpose of this guideline is to establish what data will be included in analysis and reporting of response time intervals and what data will be excluded. This process is not an exact science but is used to capture as much accurate data as possible.

Process: In the interest of consistency, a two-part filtering process will be used to establish upper and lower thresholds. The first step is the application of fixed upper and lower “macro” filters to eliminate those data points that demonstrate a clear deviation from industry and local norms. This filter is also necessary to ensure values included for analysis are numerical and are not negative numbers. These “macro” filter values are illustrated in the table below.

All Risk Classes/Categories	Definition	Lower Macro Filter	Upper Macro Filter
Alarm Processing	Time Dispatched minus Time Start	00:00:01	00:06:00
Turnout	Time Enroute minus Time Dispatched	00:00:01	00:06:00
Travel	Time Arrived minus Time Enroute	00:00:01	01:00:00
Response	Time Arrived minus Time Start	00:00:01	01:10:00
Committed	Time Clear minus Time Dispatched	00:00:01	10:00:00

Upon filtering of the dataset using the “macro” filter(s), the remaining data will then be processed using Microsoft Excel to establish the following:

Value	Microsoft Excel Function
Count	=COUNT(
Maximum	=MAX(
Minimum	=MIN(

Range	Maximum minus Minimum
Average	=AVERAGE(
Standard Deviation	=STDEV(

Upon establishing the average and standard deviation of the remaining dataset (after the application of the upper and lower “macro” filter(s)) the upper threshold will be established by multiplying the standard deviation by three (3) and adding it to the average. The lower threshold will remain at one second (00:00:01). The upper threshold is the highest value that will be included in the analysis. All values that are above the established upper threshold will be excluded from analysis. All values equal to or less than the upper threshold will be included in the analysis, down to the lower threshold. The lower threshold is the lowest value that will be included in the analysis. All values below the lower threshold will be excluded from analysis. Other situations in which call data will be excluded from analysis is listed below:

All apparatus responding in a non-emergent status
Responses outside MRFD jurisdiction (aid given)
Response data for agencies providing automatic/mutual aid to MRFD

Data will be classified and categorized based on the methodology adopted in the most recent MRFD community risk assessment-standard of cover. Response time intervals will be reported, at the 90th percentile, for each risk classification/category, per CPSE/CFAI standards. Microsoft Excel will be utilized to calculate the 90th percentile value (=PERCENTILE.INC(array,.9)).

“Rural” and “Urban” designations for CPSE/CFAI reporting purposes will be assigned based on the population density of the census block in which the call for services takes place. Those calls for service in census blocks with a population density greater than 262 people per square mile (as determined by Missoula County GIS) will be considered “urban”, whereas those calls for service in census blocks with a population density fewer than 262 people per square mile will be considered “rural”.

** This SOG supersedes any and all previous SOGs regarding this subject, known or unknown, unless prohibited by law.*

Approved: 
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